Ambient Light Compensation Through Adaptive Visual Modeling

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Abstract. *Ambient light can significantly degrade picture quality by reducing visibility of details and contrast. In many cases, it is difficult to control or eliminate ambient light, making it essential to compensate for its effect. This is especially true in the context of media and entertainment, where viewers often consume content in different lighting conditions and environments. Traditional image processing techniques, such as adjusting the backlight, brightness/contrast, or more professionally using a PLUGE signal, do not sufficiently model the human visual system. The result is that detail is lost under changes in ambient light. This is especially problematic for very dark scenes, which has become apparent with cinematic high dynamic range content where numerous consumer complaints have graced headline news.*

This paper proposes a method of ambient light compensation by adaptively modeling the contrast sensitivity functions of the human visual system. By estimating global eye adaptation to both the content and the surround environment, the contrast sensitivity models may be used to maintain perceptual detail and contrast under varying ambient illumination. We build upon existing models of human vision and show how we can use cone sensitivity to maintain global contrast. Since displays have limited dynamic range, we show how this can be adaptive to both the content and environment.

Keywords. *Ambient, light, contrast, perception, media*

Introduction

Media often serves as a focal point for social gatherings. Whether it's watching a major sporting event with friends, hosting a movie night, or a video call with distant family members, displays play a crucial role in facilitating shared experience. This versatility means that displays are used in a wide range of environments, from dimly lit livingrooms to brightly sunlit spaces.

Most professional content, however, is intended to be viewed in a very specific dim environment. ITU-R Recommendation BT.2100 [1], for example, specifies a surround mastering environment of only 5 cd/m^2. In a limited survey the BBC conducted in 2020 [2], they found that living rooms had a wide range of illuminance values and suggest that TVs support ambient compensation from 5 to at least 250 lux. The wide variance in environmental conditions is being recognized across the industry with numerous TV manufacturers implementing ambient light compensation methods. Even Hollywood's Film Maker Mode in phase 2 development is working towards a method for ambient light compensation [3].

Towards this goal, this paper investigates a method of ambient light compensation to maintain perceived detail in a wide variety of ambient light conditions. It does so based on models of the human visual system, building on both existing work, and presenting beta results of a new study.

Background

Ambient Light Compensation Methods

Ambient light in the consumer and professional space has been dealt with in a number of ways. Solutions have ranged from simply turning up the backlight, to more directly adjusting black detail with systems such as a picture line up generation equiptment (PLUGE) [4]. As display capabilities have increased with high dynamic range and faster processing, more complex methods can now be deployed. Some works have proposed methods of image enhancement that also compensate for ambient light. Kim [5] modeled contrast sensitivity under ambient light and used an adaptive spatial filter in the frequency domain. Su [6] proposed an exponential function to enhance luminance under ambient conditions with reflections. Song [7] proposed image enhancement tone mapping operators that compensate for ambient light.

This paper builds upon their work, also modeling changes in contrast sensitivity under ambient light conditions. It introduces results from a new study at lower luminance levels which improves compensation in very dark content. This method is independent from content enhancement, with the goal to simply preserve perceived image detail.

Human Visual System Cone Response

The response of the cones in the human visual system (HVS) undergoing light adaptation (L_A) is often modeled using the Naka-Rushton equation shown in equation 1.

$$R = \frac{L^n}{L^n + \sigma(L_A)^n} \qquad (1)$$

This produces a sigmoid shaped response to log luminance as show in figure 1. As light adaptation changes, the human visual system's response shifts [8]. Hence the same luminance can result in wildly different responses depending on the current state of adaptation. At a simplistic level, a just-noticeable-difference (JND) threshold between two stimulai is equivalent to a delta in cone response, ΔR. A JND is defined by 50% probability of detection. As adaptation shifts, to achieve the same ΔR, the required ΔL will change.

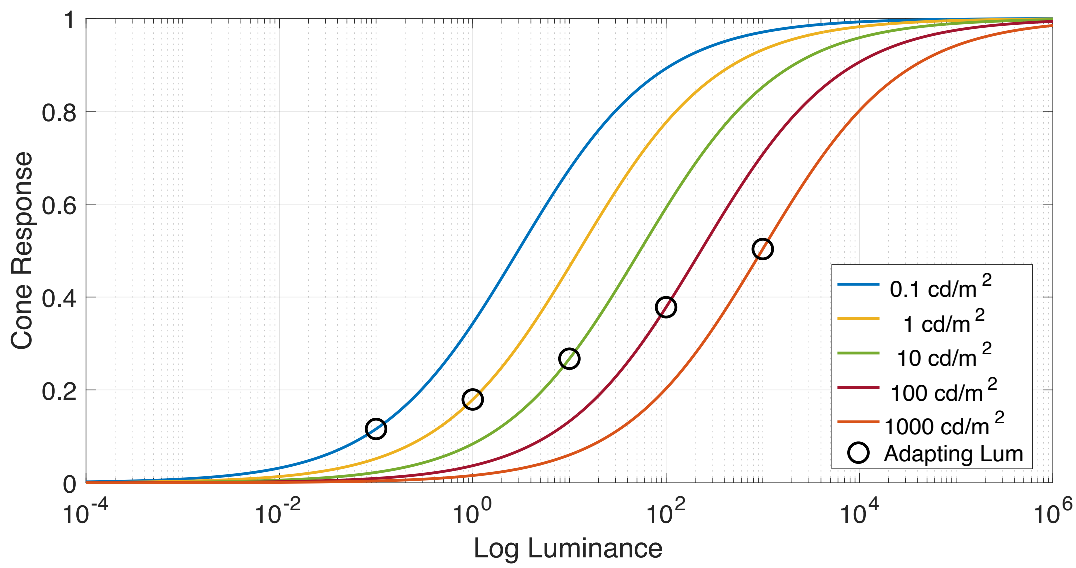

Figure 1. Cone response of the human visual system

Contrast Sensitivity

The cone response does not account for changes in sensitivity due to changes in spatial frequency. The impact of spatial frequency on JND's, where the adapting luminance is equal to the test luminance, is often measured and modeled using a contrast sensitivity function. The modulation between two stimulai is defined in equation 2 and the contrast sensitivity is inversely proportional to modulation.

$$M = \frac{L_{max} - L_{min}}{L_{max} + L_{min}} \quad (2)$$

$$S = \frac{1}{M} \quad (3)$$

Example contrast sensitivity functions (CSF) for various adapting luminance levels are given in figure 2. As the spatial frequency changes, the sensitivity required for detection changes. The peak of the sensitivity is the frequency at which the HVS is most sensitive to changes in that luminance. As the adapting luminance changes, both the overall sensitivity and the peak frequency of sensitivity changes. For the lower luminance levels, humans are most sensitive to changes at lower frequency for example.

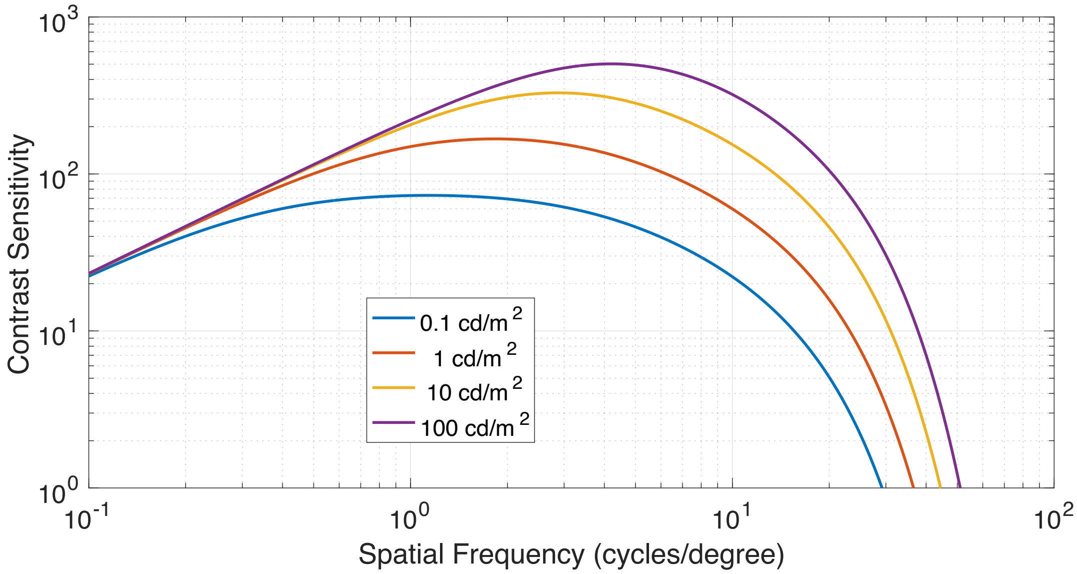

Figure 2. Contrast sensitivity of the human visual system

A widely used model for contrast sensitivity is a model developed by Barten [9] (equation 4) and is adaptive based on: field size (X) in square degrees, luminance (L) in cd/m^2, and spatial frequency (u) in cycles/degree.

$$S(u) = \frac{5200\, e^{-0.0016\, u^2 \left(1+\frac{100}{L}\right)^{0.08}}}{\sqrt{\left(1+\frac{144}{X^2}+0.64\, u^2\right)\left(\frac{63}{L^{0.83}}+\frac{1}{1-e^{-0.02u^2}}\right)}} \tag{4}$$

To accurately model contrast sensitivity, a phsychophysical experiment was conducted by Barten where observers were presented with a sinusoidal grating pattern as shown in figure 3 surrounded by a luminance the same as that of the test luminance. This way, participants were adapted to the luminance under test and were at their most sensitive state. The detectability of the sinusoidal pattern was investigated at various amplitudes. The amplitude corresponding to 50% probability of detection is the JND and the contrast sensitivity for the given adapting luminance and frequency.

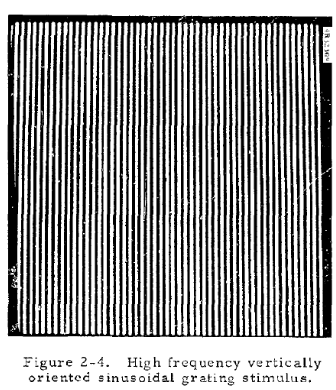

Figure 3. Sine wave pattern used in contrast sensitivity modeling

The contrast sensitivity function, and hence modulation, can therefore be used at a given luminance level and frequency to find the luminance level step that will result in a JND. This is the basis for the derivation of the Perceptual Quantization curve for HDR imagery.

Perceptual Quantizer

The Perceptual Quantizer curve (PQ), standardized by SMPTE as SMPTE ST 2084 [10], is an electro-optical transfer function (EOTF) for high dynamic range imagery that defines the translation between code words and display luminance levels. The design of PQ is such that the difference between two code values should "just barely" be invisible (to avoid contouring artifacts while still being an efficient encoding system). PQ models the JND thresholds following the Barten contrast sensitivity functions. To derive PQ, we do the following:

1. Choose a starting luminance L_i
2. Set the adapting luminance to be equal to the test luminance.

$$L_A = L_i \tag{5}$$

3. At that luminance L_i, using the Barten CSF from equation 4, find the peak sensitivity (we do this to avoid worst case scenarios) as shown in figure 4.

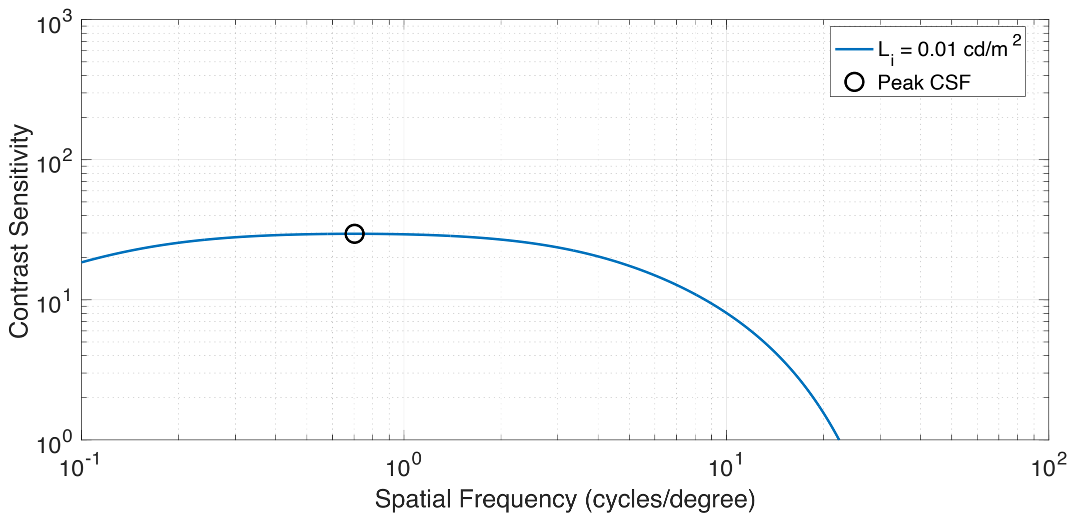

Figure 4. Contrast sensitivity curve showing the peak sensitivity (black circle)

4. From the sensitivity, using the modulation function from equation 2, we can rearrange to equation 6 to solve for the next required luminance to achieve a JND, L_{i+1} (note: in practice it is useful to take slightly below this value since a JND is still a 50% probability of detection).

$$L_{i+1} = L_i \frac{1+M}{1-M} \tag{6}$$

5. Continue the loop, incrementing i, repeating steps 2-4 to form a curve.

Each of these steps is now perceptually spaced and each code value is a JND step from the next. This forms the basis for the PQ curve as shown in figure 5.

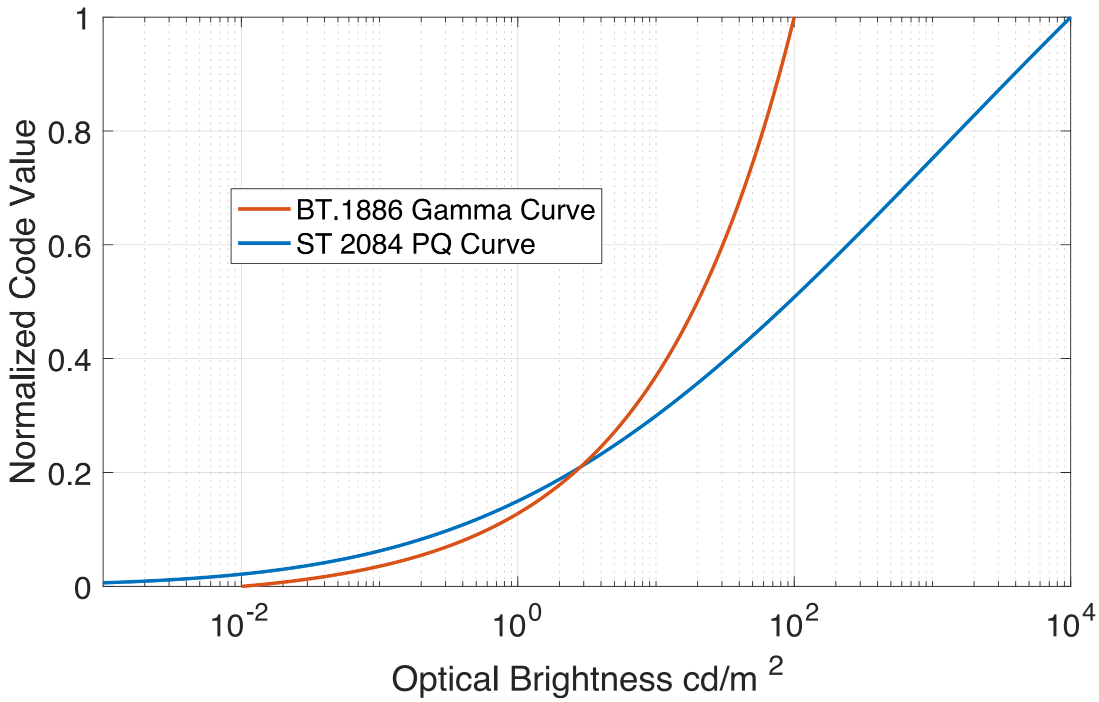

Figure 5. The Perceptual Quantizer (PQ) HDR curve compared to the standard dynamic range ITU-R Recommendation BT.1886 curve

Adapting Contrast Sensitivity for Ambient Light

The Barten CSF is designed for when the adapting luminance is equal to the test luminance. For this paper, we are interested in conditions where participants are not adapted to the luminance under test, as would be the case with ambient light. In this instance, Barten's results will need to be re-evaluated under different surround conditions.

In 1973, Rogers and Carel [11] published in a military report: "A study to determine a how surround illumination affects the contrast sensitivity". The results are somewhat limited, using only a 4˚ x 4˚ waffle grating with spatial frequencies starting from 4 cycles/degree. However, the results are still useful. In 2004, Barten designed a fit to Rogers and Carel's experiment, reformatted to be a correction factor (equation 7) to his traditional sensitivity equation.

$$f(L, L_s) = e^{-\frac{\ln^2\left(\frac{L_s}{L}\left(1+\frac{144}{X^2}\right)^{0.25}\right)-\ln^2\left(\left(1+\frac{144}{X^2}\right)^{0.25}\right)}{2\ln^2(32)}} \quad (7)$$

The reformatted results from the Rogers and Carel experiment (plotted in figure 6) show how sensitivity decreases (correction factor less than 1) as the difference between the luminance of the test and the surround increases. There is also a slight increase in the correction factor when the object luminance level is lowered, despite having the same surround ratio.

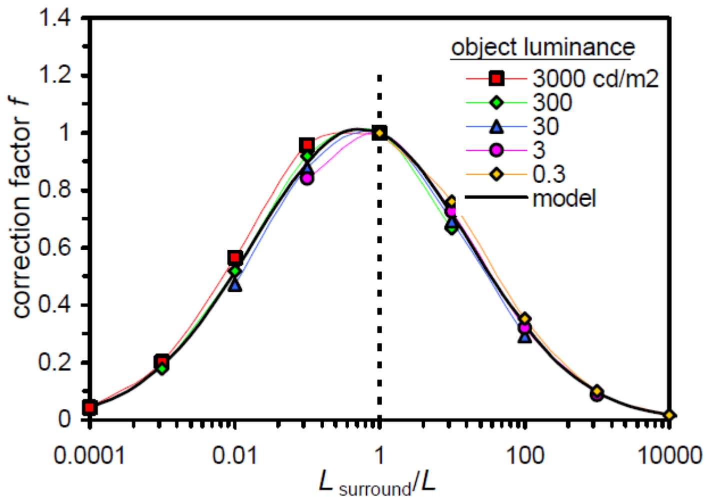

Figure 6. Correction factor for the contrast sensitivity from Rogers et al. and Barten

Theory and Problem Statement

The PQ curve used contrast sensitivity to perceptually evenly space luminance levels. Rogers and Carel propose adjustments to this contrast sensitivity under changes in adapting luminance. Using these two features together, we should be able to derive an adjustment that has perceptually evenly spaced luminance levels, under changes in adapting luminance following a similar derivation to PQ.

1. Choose a starting luminance L_i
2. Set the adapting luminance to the surround condition. We are only concerned with a brighter surround condition, hence taking the maximum between the current point Li and the surround.

$$L_A = L_S = \max\,(L_i\,, surround) \tag{8}$$

3. At that luminance L_i, using the Barten CSF from equation 4, find the peak sensitivity (we do this to avoid worst case scenarios).
4. Scale the sensitivity using the adjustment factor from equation 7.

$$M = \frac{1}{S*f} \tag{9}$$

5. From the sensitivity, using the modulation function from equation 2, we can rearrange to equation 10 to solve for the next required luminance to achieve a JND, L_{i+1} (note: in practice it is useful to take slightly below this value since a JND is still a 50% probability of detection).

$$L_{i+1} = L_{min}\frac{1+M}{1-M} \tag{10}$$

6. Continue the loop, incrementing i, repeating steps 2-5 to form a curve.

The curve developed from this procedure has equal JND steps in an ambient surround environment. With PQ, we already have equal steps in a reference environment. So, the relationship between these two curves tells us how we must adjust the image to maintain perceived detail under ambient light conditions. The result of this are shown in figure 7. On the x-axis we have the derivation of traditional PQ, when the adapting luminance is equal to the test luminance. On the y-axis we have the derivation under ambient light, when the adapting luminance is equal to the surround environment.

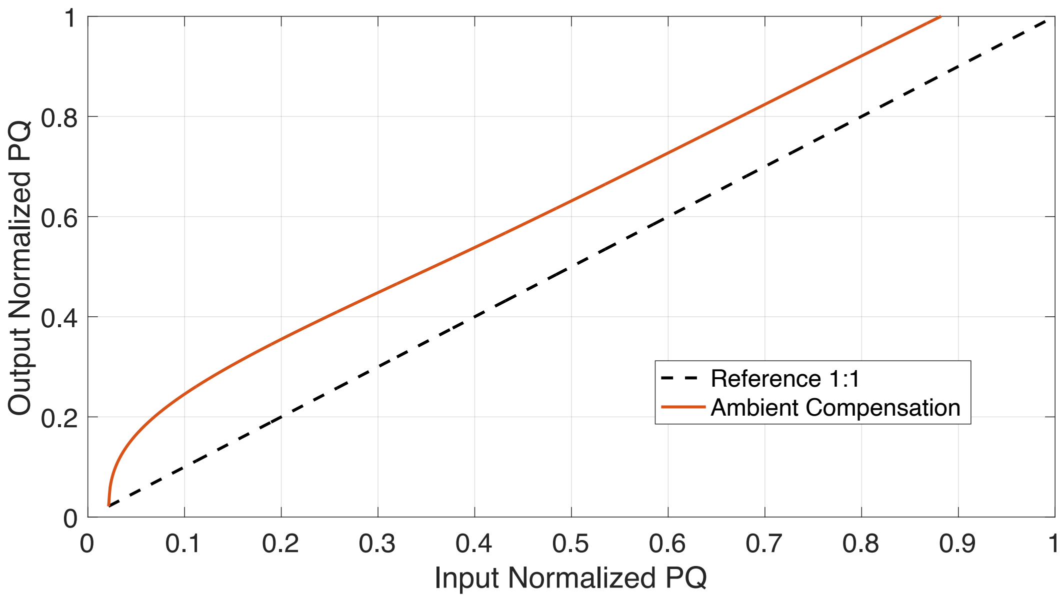

Figure 7. Changing the adapting luminance for ambient light

At the lowest luminance levels there is a substantive increase in contrast to maintain the perceived detail. Whereas, at the higher luminance levels, where the surround luminance approaches the luminance of the test condition, the slope approaches unity. Since dark details are the hardest hit in the presence of ambient light, this relationship aligns with consumer experience.

While this procedure is sound, there are a few notable limitations. First, the data from Rogers and Carel was limited to a minimum of 0.3 cd/m^2. However, in Hollywood content, especially with high dynamic range, we often have values well below this. Second, if the correction factor is too low and modulation increases above one, equation 10 will result in an unrealistically large L_{max}. Third, the compensation generally makes the image brighter. This could be ok for dark content, but would result in clipping for content that already used the full capability of the display.

Experiment and Results

In the presence of ambient light in media and entertainment, the most affected levels are very low luminance values. To build upon the data by Rogers and Carel for this region and to help avoid some of the limitations of the previous design as well as build upon other research, we mimicked their experiment setup while using significantly lower stimulus values.

Surround Levels	5, 50, 200 cd/m^2
Stimulus Levels	0.01, 0.03, 0.1, 0.3 cd/m^2

Table 1. Experiment test conditions

The experiment was conducted on a Dolby Professional Reference Monitor 4220. There were five participants who each completed the experiment twice. Results from the experiment, along with the original Barten fit to the Rogers and Carel data is shown in figure 8. At the low luminance values (high ratio), we find the experimental results show higher scalars than the Rogers and Carel data. This follows the trend from the Rogers and Carel dataset, where lower luminance stimulai resulted in higher scalar values, although to a greater extent. A fit is modeled to the experiment data for these low luminance values. The best fit is given in equation 11.

$$f = 0.93e^{-ln^3(\frac{Ls}{L})/_{155}} + 0.07 \quad (11)$$

Notice that as Ls/L goes to infinity the minimum scalar is now 0.07. This higher value reduces the likelihood of out of range values due to modulation greater than one. The optimized function is shown as blue dashed lines in figure 8 with standard-error error bars. The original Barten fit is shown as black dashed lines.

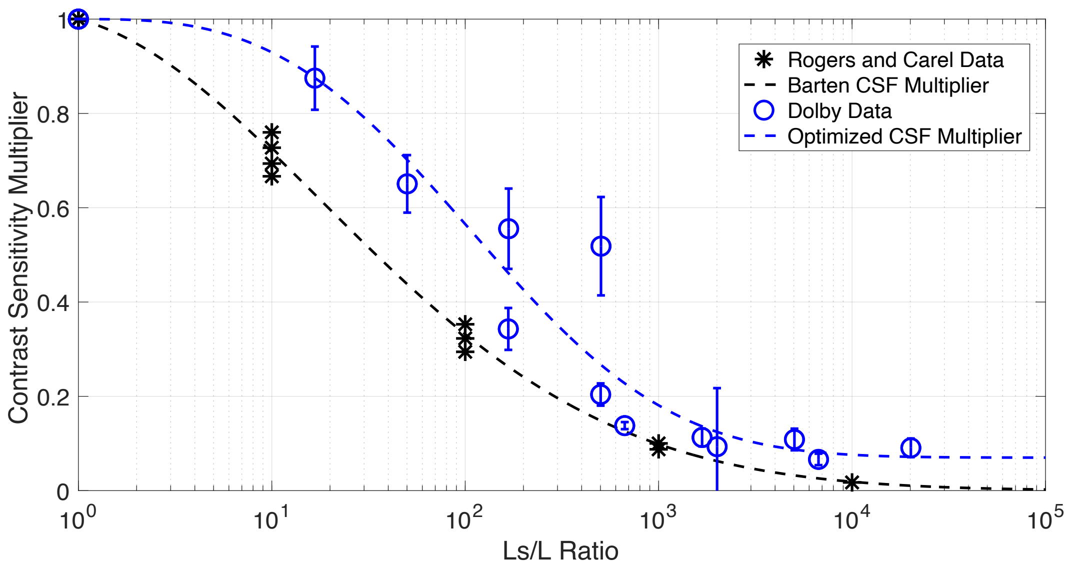

Figure 8. CSF multiplier results for ultra low luminance stimulai

Algorithm Design

With procedure basics in place, and adjustment/complementary studies completed, we can design the final system. Since the derivation is based on PQ, there are shortcuts we can take to avoid the direct and costly optimization on the contrast sensitivity functions themselves. The block diagram is given in figure 9.

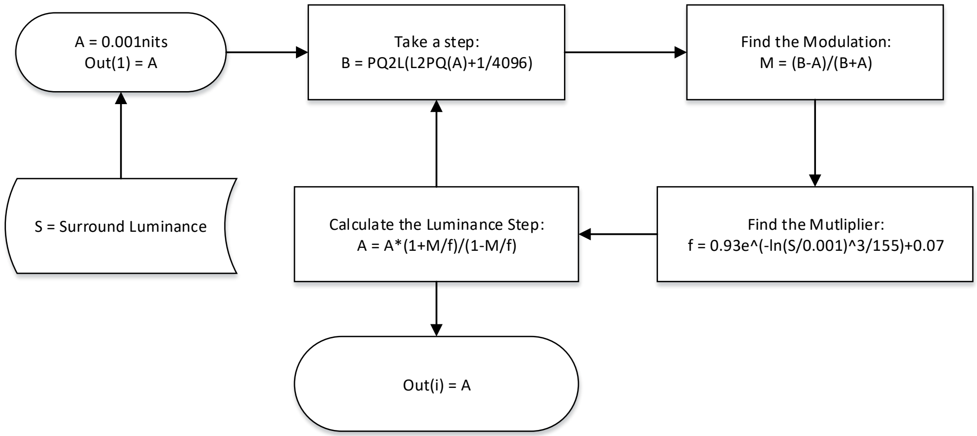

Figure 9. Block diagram of the adjustment curve derivation

First pick a starting luminance value. In the case of figure 9 we start with 0.001 cd/m^2. The standardized PQ curve can be used to determine the modulation required for the next JND by converting from luminance to code word, taking a 12-bit (1/4095) step, and converting back to luminance. Using the new equation for the adjustment multiplier, we can find the new luminance for a JND. Storing these values and iterating results in the compensation curves. A look-up-table (LUT) interpolation approach can be used to process the adjustment in real-time as shown in equation 12, where *L* is the input luminance value and PQ_{orig} and PQ_{amb} are the curves created without and with ambient compensation.

$$Q(L) = LUT(L, PQ_{orig}, PQ_{amb}) \tag{12}$$

The last stage is mapping the curve range to avoid clipping. There are many ways to design this process including dyanamic methods, such as those in [7]. Dark images, for example may not require a normalization roll-off curve. For processing simplicity, in this paper we will consider a static normalization shown in equation 13, where *a* is the minimum of the source container and *b* is the maximum of the source container.

$$y(PQ_{amb}) = \frac{PQ_{amb} - Q(a)}{Q(b) - Q(a)} * (b - a) + a \tag{13}$$

In this paper, we will aim to keep images within the PQ container range 0-10,000 cd/m^2. So *a* will be 0 and *b* will be 10,000. Normalization in a JND space such as PQ will give improved perceptual results. The resulting curves for ambient compensation are shown in figure 10. Example processed images are shown in figure 11.

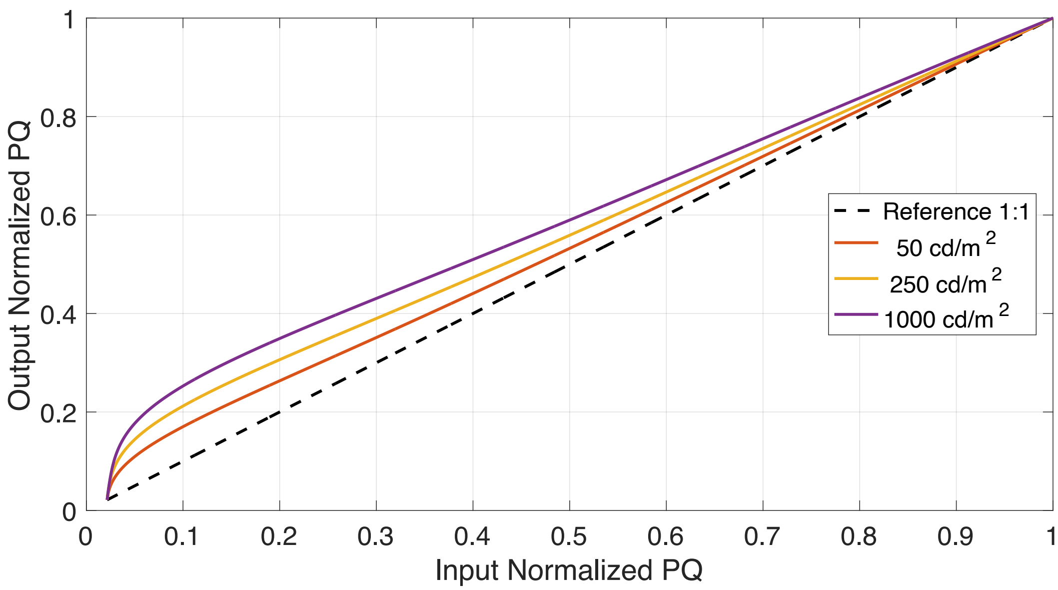

Figure 10. Ambient compensation curves

a b c

Figure 11. Images with ambient compensation (a) original, (b) 100cd/m^2, (c) 500cd/m^2

Conclusions

In media and entertainment, ambient illumination can significantly degrade the visibility of detail. In this paper we have presented a method of ambient compensation, using models of the human visual system, that allow for maintaining perceived detail and contrast. We presented results of a new study that extend validity of ambient compensation to low luminance values common in cinematic content. Future work includes studying dynamics of the post compensation normalization, spatial localization, and compensation for specular reflections.

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